

Multicopter Physical Specifications Questionnaire

For each airframe in the `_Tuned.af` files, please provide the following physical specifications so we can accurately model the dynamics in simulation.

Required Parameters

Parameter	Unit	Description	Example
PHYS_MASS_KG	kg	Total AUW (All-Up Weight) including battery	0.91
PHYS_FRAME_MM	mm	Diagonal motor-to-motor distance	220
PHYS_MOTOR_KV	RPM/V	Motor KV rating	1000
PHYS_PROP_DIAMETER_INCH	inch	Propeller diameter	11
PHYS_PROP_PITCH_INCH	inch	Propeller pitch	3.7
PHYS_PROP_BLADES	count	Number of blades per prop (2, 3, 4)	2
PHYS_CELL_COUNT	S	Battery cell count (1S, 2S, 3S, 4S, 6S)	4
PHYS_BATTERY_MAH	mAh	Battery capacity	1500
PHYS_ARM_LEN_M	m	Center-to-motor distance	0.11
PHYS_INERTIA_ROLL_PITCH	kg·m ²	Roll/Pitch moment of inertia	0.002
PHYS_INERTIA_YAW	kg·m ²	Yaw moment of inertia	0.005
PHYS_MAX_THRUST_N	N	Max thrust per motor (measured or estimated)	3.5

Optional (Fixed Wing Only)

Parameter	Unit	Description
PHYS_CRUISE_SPEED_MS	m/s	Cruise airspeed
PHYS_MIN_SPEED_MS	m/s	Minimum airspeed (stall)
PHYS_MAX_SPEED_MS	m/s	Maximum airspeed
PHYS_LIFT_COEFF	-	Wing lift coefficient
PHYS_DRAG_COEFF	-	Parasite drag coefficient

Airframes Needing Specs

Ecks Series (all 2208 1000kv, 11x3.7" 2-blade, 4S)

- ☐ **EcksTuned.af** — 910g AUW, 220mm frame
- ☐ **Ecks_220mm_Tuned.af** — same as above
- ☐ **Ecks_800g_Moderate.af** — 800g AUW, 220mm frame

- ☐ **Ecks_800g_Sport.af** — 800g AUW, 220mm frame
- ☐ **Ecks_1kg_Moderate.af** — 1kg AUW, BR2205 2300kv + Cyclone 5x3x3
- ☐ **Ecks_1kg_Sport.af** — 1kg AUW, BR2205 2300kv + Cyclone 5x3x3
- ☐ **EcksQuatQ.af** — quaternion retune

Ken's Models

- ☐ **Ken_s_Alpha_Test_Tuned.af** — LOW_VOLT_THRES=13.8V (4S?), 450mm frame?
- ☐ **Ken_s_450_1165_Tuned.af** — 450-1165 frame, what motors/props?
- ☐ **Ken_s_LadyBug_Tuned.af** — WLToys brushed 1S, what frame/motors?

Other

- ☐ **DevEBox_Tuned.af** — what hardware?
 - ☐ **S500_1137_Tuned.af** — S500 frame, what motors/props?
 - ☐ **150mm_Brushed_Tuned.af** — 150mm brushed, 1S/2S?
 - ☐ **Rok_s_Quad_Tuned.af** — AF_TYPE=QUAD (not QUAD_X), LOW_VOLT_THRES=10V (3S?), BATTERY_CAPACITY=50 (5000mAh?), ESC=FAST_PWM, what frame/motors/props?
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How to Fill

1. **Mass:** Weigh with battery installed (AUW)
 2. **Frame:** Measure motor-to-motor diagonal
 3. **Motor:** Check label or invoice for KV
 4. **Prop:** Check prop markings (e.g., "11x3.7" = 11" dia, 3.7" pitch)
 5. **Battery:** Check label for S-count and mAh
 6. **Arm length:** Half of frame diagonal (or measure center to motor)
 7. **Inertia:** If unknown, use approximation:
 - $I_{roll_pitch} = \text{mass} \times (\text{arm_len})^2 / 12$
 - $I_{yaw} = 2.5 \times I_{roll_pitch}$
 8. **Max thrust:** If unknown, estimate from motor+prop data or use $(\text{mass} \times 9.81) / (\text{hover_throttle} \times \text{num_motors})$
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Output Format

Please provide as:

EcksTuned.af:

```
PHYS_MASS_KG = 0.91
PHYS_FRAME_MM = 220
PHYS_MOTOR_KV = 1000
PHYS_PROP_DIAMETER_INCH = 11
PHYS_PROP_PITCH_INCH = 3.7
PHYS_PROP_BLADES = 2
PHYS_CELL_COUNT = 4
PHYS_BATTERY_MAH = 1500
PHYS_ARM_LEN_M = 0.11
PHYS_INERTIA_ROLL_PITCH = 0.002
PHYS_INERTIA_YAW = 0.005
PHYS_MAX_THRUST_N = 3.5
```

Save responses to: uavx-python/src/airframes/SPECS_RESPONSES.md